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Segmented Animation, User-Control Strategy and Cognition

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ABSTRACT

The aim of this study was to investigate the effects of various user-control strategies of segmented animation in learning abstract contents or processes that are not naturally visual on the cognition of students. In particular, the study examined the effects of five different user-control strategies: linear user-control, random user-control, free user-control, program-control, and continuous user-control. The research design was quasi-experimental and the data obtained were statistically analysed using ANCOVA and ANOVA. The instruments involved were pre-test, post-test, and NASA-TLX cognitive load test. The sample size consisted of 265 semester-two students enrolled in the Diploma in Networking System. The results indicated significant differences in the post-test and cognitive load test outcomes. In conclusion, this study suggests that the use of user-control, either linear user-control or random user-control, were sufficient strategies for learning abstract contents in segmented animation form.

1. Introduction

Presently, animation has become a preferred instructional tool in teaching and learning processes (Ainsworth, 2008; Aldalalah, Fong, & Ababneh, 2010; Ayres & Paas, 2007; Chandler, 2009; Fong & Lily, 2010; Mayer, 2005). Animation is defined as a sequence of fast-changing screen displays that represent the movement or changes from one dimension to another (Ainsworth, 2008; Lightner, 2001; Mayer & Moreno, 2002). Thus, animation could play a potential role in reinforcing the understanding of dynamic processes, such as those not viewable in real space and time scales (e.g.: atom formation), real processes that are practically impossible to be exposed in a learning setting (e.g.: hydroelectric turbine function), or processes that are not naturally visual (e.g.: networking protocol). Furthermore, animation plays an important role in minimizing the cognitive cost of mental simulation (Narayanan & Hegarty, 2002). Consequently, this will save cognitive resources for learning tasks, particularly for novices and students with poor imagery or visualization abilities (Betrancourt, 2005). In its best uses, animation presents information in a more understandable and memorable way than static visual presentation (Boucheix & Schneider, 2009; Hegarty, Kriz, & Cate, 2003; Lewalter, 2003). Though, even with these advantages and theoretical support, the findings of comparative studies between animation and static media remain inconsistent (Ainsworth, 2008; Ayres, Marcus, Chan, & Qian, 2009; Hegarty, 2004; Lin & Dwyer, 2010, 2004; Sperling, Seyedmonir, Aleksic, & Meadows, 2003). The core causes of inconsistencies might be mainly due to design attributes (Ahmad Zamzuri, 2013; Tversky, Morrison, & Betrancourt, 2002). This mostly happens due to the instructional designers' failure in addressing the aspects related to students' cognitive abilities in their design and development process (Jeroen, Enboer, & Sweller, 2005; Lowe, 2004). Students' cognitive abilities actually could be improved by imposing adequate design features, such as simple and uncluttered visuals, uncomplicated content, user-friendly interaction, adequate speed of animation display, and type of user-control. However, it is difficult to determine the ideal speed and level of user-control due to the variation of students' learning ability and style. Nevertheless, utilizing segmentation principles could be a potential solution in addressing this problem (Mayer & Chandler, 2001; Mayer & Moreno, 2003; Moreno, 2007).

1.1. Animation and cognition

Learning is a process of receiving, coding, and storing information, in addition to the ability to retrieve the stored information appropriately (Lin & Dwyer, 2004). Basically, human memory is divided into three main components which are sensory memory, short-term or working memory and long-term memory (Atkinson & Shiffrin, 1971, 1968). Information normally stored in the sensory memory takes nearly 0.5 to 1 s, while working memory will store the information in around 5 to 15 s (Atkinson & Shiffrin, 1971). Earliest information enters the working memory through the sensory memory will be stored within this time frame until the following information is received. According to Norman (1982), the first information will expire exponentially from memory, that is 1/3 of information is lost for every 100 to 150 milliseconds, as depicted in Figure 1. This limitation, under some conditions, will cause some information entered into the

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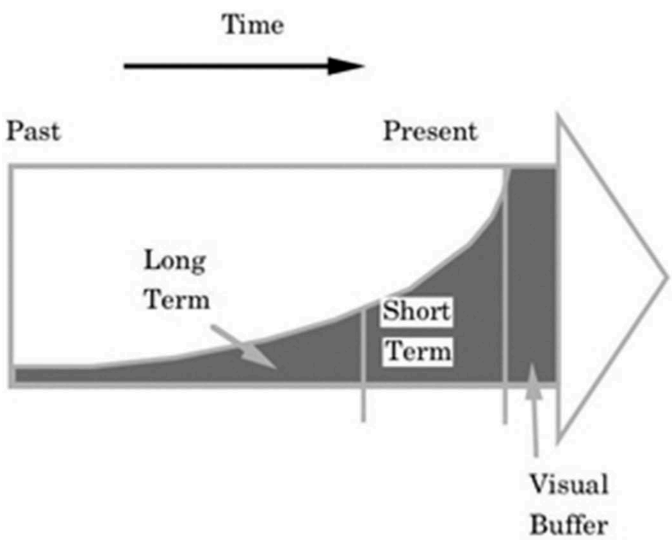

Figure 1. Information structure in memory.
(Source: Wiebe, 1991)

working memory to not be registered permanently in the long-term memory (Clark, Nguyen, & Sweller, 2011; Jun-Xia, 2007; Sweller, 2004). For instance, information in an animation form could contribute to the inability of the working memory to accommodate continuous information flow due to the limited process duration and limited storage capacity (Wiebe, 1991). The failure in registering sufficient amounts of information in the long-term memory will surely be an influential factor contributing to learning failure (Janssen, Erkens, Kirschner, & Kanselaar, 2010; Paas, van Gog, & Sweller, 2010; van Merriënboer & Ayres, 2005).

Next, the information which enters the long-term memory will be structured and organized into the schemas form (Paas, Renkl, & Sweller, 2003; Sweller, 2004; van Merriënboer & Ayres, 2005; van Merriënboer & Sweller, 2005). According to cognitive theory, learning is a process of integrating new information with existing schemas for new schema formation (Kalyuga, Ayres, Chandler, & Sweller, 2003; Mayer, 2002; Paas et al., 2003). Thus, the ability to understand and recall information to be registered in the long-term memory also depends on the activation of existing schemas (Mayer, 2002; Paas et al., 2003; Sweller, 2004). Accordingly, the appropriate combination of technology and pedagogy is essential to stimulate students' ability in activating the existing schemas to understand the new information received (Ayres & Paas, 2007; Lowe & Boucheix, 2010). Previous studies on cognition indicate that conventional teaching and learning modes do not critically consider these memory structure limitation issues (Tindall-Ford, 1998).

Animation seems to have a strong theoretical foundation in fulfilling the requirements of cognitive theory, as discussed above. This is evident through the multimedia learning theories, such as Paivio's dual-coding theory (Paivio, 1986) and Mayer's cognitive theory of multimedia learning (Mayer, 2001). According to these theories, information is processed in the human's brain through two channels, namely the verbal channel and the visual channel. Activation of both channels through the referential process can be utilized for meaningful schema formation in the long-term memory throughout the learning process (Mayer, 2001; Paivio, 1986). Therefore, animated instructions which are developed with a combination of various interconnected multimedia elements can serve as an effective tool for successful learning. Thus, the right strategy and approach of verbal and visual information presentation is essential to ensure the meaningful schema formation in the long-term memory. Given the limitation of working memory in term of duration and capacity, an inadequate strategy will lead to increased cognitive load throughout the dual-coding activation in the learning process (Sweller, 1994, 1988). Referring to the cognitive load theory, there are three categories of cognitive load that affect the ability of working memory, namely the intrinsic cognitive load, extraneous cognitive load and germane cognitive load (Kolloffel, Eysink, De Jong, & Wilhelm, 2009; Sweller, van Merriënboer & Paas, 1998).

Basically, the intrinsic cognitive load happens due to the inherent level of difficulty associated with the instructional content (Chandler & Sweller, 1991). Nevertheless, as previously discussed, presenting information in an animated form could possibly minimize this load, specifically in depicting dynamic processes. For long and complicated animations, chunking the animation into segments and presenting them in isolation, appeared to be a potential solution in addressing the inherent difficulty (Kirschner, Sweller, & Clark, 2006). Meanwhile, the extraneous cognitive load is primarily attributed to the instructional media design techniques and delivery strategies (Chandler & Sweller, 1991, 1992). In order to minimize this load, it is important for the instructional designers to closely follow the research-grounded multimedia principles which have been established in the field of User Interaction (UI). This is to ensure that effective interaction occurs between the user and instructional material for successful learning. Referring to the segmented animation principle, an ideal level of user-control and interaction is crucial to be studied to lessen the extraneous cognitive load. Therefore, the focus of this study will mainly be on identifying the ideal user-control strategy of segmented animation in learning abstract contents. Lastly, the germane cognitive load is the load associated with the processing, construction, and automation of schemas (Sweller, van Merriënboer, & Paas, 1998). This has been known as the most efficient and effective cognitive load in the context of instructional media (Gerjets, Scheiter, & Schorr, 2003). It is suggested that by minimizing the extraneous cognitive load, it will promote the germane cognitive load (Sweller et al., 1998). Accordingly, it is vital for instructional designers to minimize the extraneous cognitive load and redirect students' attention to cognitive processes that are exactly related to the formation of schemas (Sweller et al., 1998).

1.2. Segmentation and user-control

Speed and visual complexity appear to be the likely causes that impede students' understanding of the contents presented in an animation form (Lowe, 2004, 1999; Tversky et al., 2002; Weir & Heeps, 2003). This is due to their failure in extracting and processing necessary, meaningful information from the continuous information flow of an animation in the memory structure (Ayres et al., 2009; Lowe, 2004, 1999; van Oostendorp, Beijersbergen, & Solaimani, 2008; Weir & Heeps, 2003).

Therefore, animations that were designed without addressing this limitation would probably distort students' focus and attention in learning, and thus failing its role as an instructional aid (Torres & Dwyer, 1991). Without suitable time allocation, students would not be able to build an accurate mental model for new schema formation by activating existing schemas in the long-term memory (Garhart & Hannafin, 1986). Presentation strategies that allocate ample interactive time will potentially help in processing continuous information flow in the memory structure (Slater & Dwyer, 1996). Apparently, segmented animation with user-control features that allow students to control the segments' viewing rather than continuous passive viewing may be a more effective solution for this shortcoming (Boucheix & Guignard, 2005; Mayer & Chandler, 2001; Mayer & Moreno, 2003; Moreno, 2007).

A study by Moreno (2007) on a group of prospective teachers had proved the effectiveness of segmented animation in teaching. In the study, they were taught seven basic teaching skills by an expert teacher in the form of animation and video. The research outcome revealed that students in the segmented animation mode performed better and reported lower levels of cognitive load than those in the uncontrolled continuous animation mode. This is due to the fact that the rate of information received by the working memory is sufficient and the cognitive load is at the minimum level for students in the segmented animation mode (Moreno, 2007). Nevertheless, it was not stated clearly what type of user-control was used in the segmented animation. A study by Mayer, Dow, and Mayer (2003) on the topic of how an electric motor works in a narrated animation condition also revealed similar outcomes. Both segmented animation with user-control and continuous animation without user-control were used in the study. The students in the segmented animation mode had the capability to control the movements from one segment to another according to their preference. They also could repeat the segments by utilizing the replay buttons in each segment according to need. The study's results indicated that the segmented animation group benefited more from the given learning session. It has been identified that the segmentation and user-control elements had contributed to the result of the study.

The role of user-control of segmented animation had also been studied by Boucheix and Schneider (2009). Three user-control strategies were used for learning the functioning of a three-pulley system in this study. The strategies were continuous animation with one click user-control, animation with partial user-control, and animation with full user-control. Continuous animation referred to a single continuous display without a break, which will be played when the students clicked on the illustration area. The display speed rate was relatively slow and the average duration of entire animation was 8.5 s. While, the partial user-control animation had five segments with buttons provided to control the movement from one segment to another. Students were only allowed to repeat as they finished viewing the whole segments. In the full user-control animation, every movement at the pulley system could be determined freely by using the mouse. The display speed rate could also be determined freely by the students. Overall, the results of Boucheix and Schneider (2009) revealed that students gain greater learning benefit from segmented animation with user-control than nonstop animation, specifically students with low spatial and mechanical reasoning abilities. On the other hand, control elements seem to also encourage them to spend more time browsing each segment. However, controllable animation did not have a powerful effect on learning for students with high spatial and mechanical reasoning abilities.

The findings and summary of study by Hasler, Kersten, and Sweller (2007) also proved that segmentation and user-control in animation have positive effects in learning. In Hasler et al. (2007), a comparison was made between four presentation strategies: two with user-control and narration, one continuous animation and narration, and another with narration only presentation. The content of the animation was about the causes of day and night and the audio narration were same for all four strategies. The first user-control strategy consisted of an animation which was subdivided into 11 segments. Students could view each segment in sequence, which they could then continue to the following segment by clicking the play button once the segment display was completed. In the second user-control strategy, students could decide the animation segment by using the stop-play buttons. They could stop anywhere by clicking the stop button and proceed by clicking the play button if they wished to continue from the stop point. Overall, the findings of Hasler et al. (2007) revealed that students in user-control animation mode had obtained higher test performance with relatively lower cognitive load compared to the continuous animation and narration only groups. Similar achievement has also been attained by students using stop-play animation, even though they did not regularly use the stop-play features throughout the learning process.

Spanjers, Wouters, van Gog, and van Merrienboer (2011) have also proved the effectiveness of segmented animation in learning of probability calculations. The animations used were the continuous animation and segmented animation with program-control. In the program-control, the animation was chunked into eight segments and each segment played until the segment's end and then paused for around 2 s before moving on to the following segment automatically. In the continuous animation, all eight segments were displayed continuously without any break inbetween. In this study, the groups were divided into a low prior knowledge and a high prior knowledge respectively. The overall study findings revealed that students in the program-control segmented animation mode benefited more from the learning session compared to the students in the continuous animation mode, specifically the low prior knowledge students. However, the segmented animation was not significant for the high prior knowledge students.

In conclusion, all the findings presented above proved that the segmented animations with user-control or program-control features are more effective than continuous animation. However, the question arises concerning which is the most effective form of control strategy. This is due to the fact that all the studies above focus on comparing segmented animation and continuous animation and not the user-control strategies. Thus, it is necessary to conduct studies that focus on identifying the most effective control strategy of segmented animation for successful learning. This is important since inadequate control will possibly undermine the learning effectiveness, particularly the complex and abstract instructional topics (Boucheix & Schneider, 2009). Furthermore, inadequate control will also cause students to pay unnecessary attention on the control compared to the learning contents. This was seen in the study by Hasler et al. (2007). Moreover, previous studies have mostly focused on known topics which probably help students in constructing a suitable mental-model based on existing knowledge. This may not apply for abstract topics that students have never seen before. Therefore, this study will focus on the effects of various user-control strategies of segmented animation in learning abstract contents or processes that are not naturally visual.

2. Methodology

2.1. Research objective

Based on the literature overview above, the main objective of this research was to study the effects of various user-control strategies of segmented animation on the cognition of students in learning abstract contents. In detail, the questions derived from the literature overview are as follows:

RQ1. Will there be any significant difference in post-test achievement of students who undergo different user-control strategies of segmented animation?

RQ2. Will there be any significant difference in cognitive load test outcomes of students who undergo different user-control strategies of segmented animation?

2.2. Teaching material

Five courseware prototypes with five different user-control strategies were developed for the research purpose. Each courseware contains five similar animations which depict the network protocol and troubleshooting flow. Whereby, the content is abstract and categorized as processes that are not naturally visual. Text explanations were also included as verbal support. This is to utilize both the visual and verbal channels in the memory structure, in order to reduce the load of working memory as highlighted in Mayer (2001). The animation and the text explanations are presented in separate blocks as in Figure 2. Meanwhile, the user-control buttons are placed at the bottom center of the animation block as in Figure 3. All the courseware are similar in terms of content and interface design and the only difference is the user-control strategy used.

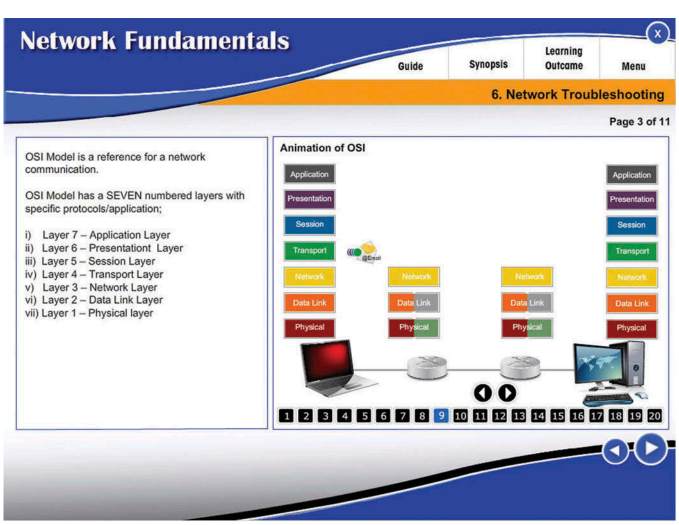

Figure 2. Courseware interface.

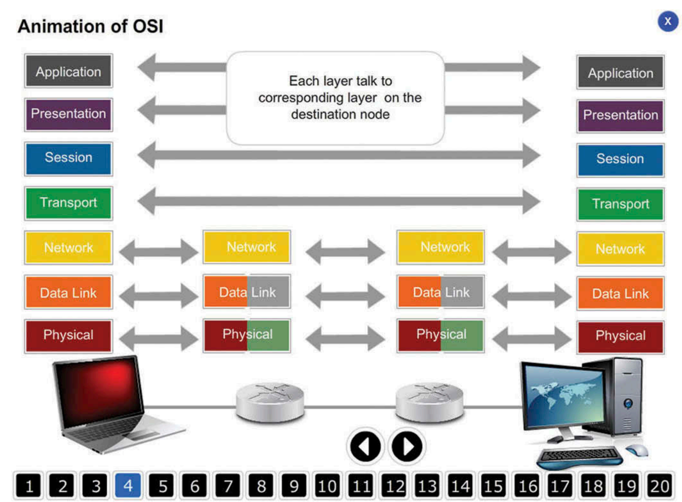

Figure 3. Animation block.

The students could also view the animations in full-screen mode by clicking on the animation block. The overall development of the animation courseware had carefully followed the multimedia principles which had been established in the field of User Interaction (UI) to ensure that the intrinsic and extraneous cognitive load would be minimized.

The first control strategy was linear user-control (LUC), which the animations were chunked into segments and the students could display the segments in linear sequence by clicking the next and previous buttons as in Figure 4.

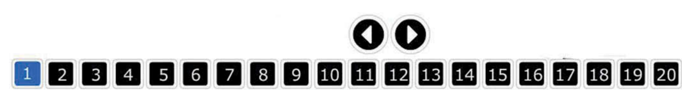

Figure 4. Linear user-control.

Only the next and previous buttons function. Meanwhile, the highlighted number indicates the segment being displayed, it is not a button.

It appears to be no well-defined guidelines or principles in the literature in determining the best segment length. However, the two options available, based on the researchers practice are:

(1) The length is based on theories with regards to cognitive functioning.
(2) The length is based on the content expert's view, in order to determine the meaningful segments (Spanjers, van Gog, & van Merrienboer, 2010).

This study implemented both options. The animation was broken into meaningful segments based on the expert's view. The segments were kept short, which was both necessary and helpful for novices, since the entire information was new for them (Spanjers et al., 2010). Basically, the animations were chunked into 20 meaningful segments. Each segment's display was approximately 10 to 12 s.

The second control strategy was random user-control (RUC), which the animations were also chunked into segments as in LUC. As in LUC, the animations were also chunked into 20 meaningful segments in RUC. Each segment's display was approximately 10 to 12 s. The main difference between LUC and RUC is students could display the segments randomly by clicking the preferred segment's button in RUC. For instance, to display segment 10 in RUC, the students just need to click button 10 as in Figure 5.

Figure 5. Random user-control.

Whereby, in LUC, the students need to click the next or previous button continuously to move from one segment to another in linear sequence, this can be skipped in RUC. In LUC, only the next and previous buttons function. Meanwhile, in RUC, all the 20 segment buttons function as in Figure 5.

The third control strategy was free user-control (FUC), which allow the students to determine the segmentation of animation based on their own preferences. This can be done by clicking the play, pause and replay buttons as in Figure 6. To start the animation display, students just need to click the play button and they could stop the animation at any point by clicking the pause button. To proceed the display from the stop point, students need to click the play button again. The replay button is to start the animation from the beginning. The concept is similar to the conventional video player applications. Therefore, the icon used for play, pause, and replay buttons are standard icons used in video players.

Figure 6. Free user-control.

The fourth control strategy was program-control (PC), which the animations were chunked into segments as in LUC and RUC. The animations will be displayed nonstop when the students click the replay button as in Figure 7. The pause rate between segments was determined by the instructional developer. Each segment's display was approximately 10 to 12 s. The pause duration between segments was 3 s, which was deemed suitable based on arguments in Ahmad Zamzuri (2013) and Spanjers et al. (2011). The segment displayed will be highlighted as in Figure 7, it is not a button. Only one button functions in PC, which is the replay button. By clicking the replay button at any point, the animation will be displayed from the beginning again.

The fifth control strategy was continuous user-control (CUC), which the animations will be displayed continuously when the students click the start button as in Figure 8. The students could replay the animations by clicking the start button when the display ended or at any point of the animation display. The main difference between CUC and PC is there is no static display for approximately 3 s in between segments in CUC.

Figure 7. Program-control.

Figure 8. Continuous user-control.

2.3. Procedure

This quasi-experimental study investigated the effects of five different user-control strategies of segmented animation on the cognition of students in learning abstract contents. The independent variable was the user-control strategy, and the dependent variable was the post-test and cognitive load test. The research participants were 265 semester-two students enrolled in the Diploma in Networking System from five different polytechnics. Two classes from each polytechnic involved in the study. The classes are not assigned randomly and have different lecturers. The study was conducted separately for all the classes in a controlled lab environment. Each polytechnic undergo different treatment respectively (i.e. different user-control strategy). Basically, they were considered to have low prior knowledge of the content presented as also identified from the pre-test mean score: LUC (*M = 22.51 SD = 9.43*), RUC (*M = 20.18, SD = 5.60*), FUC (*M = 22.57, SD = 9.44*), PC (*M = 24.55, 11.64*). and CUC (*M = 28.27, SD = 11.06*). Instruction on the courseware's features was given prior the study. Students were allocated 40 min of learning time in a controlled lab setting. Two instructors assisted in monitoring the learning session. No verbal explanations from the instructors were given throughout the study. The study was conducted separately for all groups. The pre-test was conducted before the learning process and the cognitive load test was conducted immediately after the learning process and followed by the post-test.

2.4. Test instrument

Pre-test and post-test were used on five groups that viewed five different user-control strategies respectively. Pre-test and post-test were written tests that measured all cognitive levels outlined in Bloom's Taxonomy (Bloom, 1956). The test contains nine fill-in-the-blank questions and two essay questions. 40 min was allocated for students to complete the test. ANCOVA was employed to statistically analyze the obtained data.

NASA-TLX (Hart & Staveland, 1988) was used to measure the students' cognitive load. NASA-TLX is a multi-dimensional assessment tool to obtain cognitive load via six subscales, each associated with the intrinsic and extraneous load (Hart, 2006). The subscales scores were then combined and derived into an overall weighted cognitive load score. NASA-TLX was developed by the Human Performance Group at NASA's Ames Research Center over a three-year development cycle that included more than 40 laboratory simulations (Hart & Staveland, 1988). It is a reliable tool to measure cognitive load and had been used in numerous studies in various fields since its introduction (Hart, 2006). Furthermore, NASA-TLX has been translated to more than a dozen languages and subjected to number of independent evaluations in which the reliability and utility were assessed and compared to other methods of measuring the cognitive load (Hart, 2006). ANOVA was employed to statistically analyze the obtained data.

3. Data analysis

RQ1. Will there be any significant difference in post-test achievement of students who undergo different user-control strategies of segmented animation?

A one-way ANCOVA statistical analysis was conducted to investigate if the different user-control strategies of segmented animation have a significant effect in influencing the post-test achievement. The independent variable was the segmented animation with five different user-control strategies and the dependent variable was the post-test scores conducted following the completion of the learning process. Pre-test scores conducted prior to the commencement of the learning process were used as the covariate to avoid scores on the covariate affect the treatment (Pallant, 2007). SPSS uses regression procedures to remove the variation in the dependent variable that is due to the covariate or pre-test, and then performs the normal analysis of variance techniques on the adjusted scores (Pallant, 2007).

From the one-way ANCOVA test, Levene's test for homogeneity of variances was not significant ($p > 0.05$), and therefore, the data do not violate the assumption of equality of error variances. After adjusting for pre-test scores, there was significant difference on post-test achievement, $F\ (4,259) = 5.99$, $p < 0.05$, *partial eta squared* $= 0.09$, with effect size between medium and high according to Cohen's 1998 guidelines (Pallant, 2007) as shown in Table 1.

The adjusted mean scores indicated that students in the RUC mode obtained a higher mean score ($M = 67.75$, $SE = 1.56$) than students in LUC mode ($M = 67.33$, $SE = 1.67$), PC mode ($M = 61.76$, $SE = 1.57$), FUC mode ($M = 60.71$, $SE = 1.60$) and followed by CUC mode ($M = 58.86$, $SE = 1.67$), as shown in Table 2. To further analyze the results obtained, the Bonferroni test was carried out for between groups comparison study.

The results of the Bonferroni test indicated significant difference in terms of adjusted mean scores of post-test between LUC and FUC ($MD = 6.62$, $p < 0.05$) with LUC obtaining a higher mean score, a significant difference between LUC and CUC ($MD = 8.48$, $p < 0.05$) with LUC obtaining a higher mean score, a significant difference between RUC and FUC ($MD = 7.04$, $p < 0.05$) with RUC obtaining a higher mean score, and a significant difference between RUC and CUC ($MD = 8.89$, $p < 0.05$) with RUC obtaining a higher mean score. The results of the Bonferroni test also indicated no significant difference between PC and LUC ($MD = 5.58$, $p > 0.05$), PC and RUC ($MD = 5.10$, $p > 0.05$), PC and FUC ($MD = 1.05$, $p > 0.05$), PC and CUC ($MD = 2.90$, $p > 0.05$), as well as no significant difference between FUC and CUC ($MD = 1.85$, $p > 0.05$). This showed that LUC and RUC obtained the highest mean scores and it was significantly different with PC and CUC. Meanwhile, there appears to be no significant difference in terms of mean scores between PC and other user-control strategies.

RQ2. Will there be any significant difference in cognitive load test outcomes of students who undergo different user-control strategies of segmented animation?

A one-way ANOVA statistical analysis was conducted to investigate if the different user-control strategies of segmented animation have a significant effect in influencing the cognitive load. The independent variable was the segmented animation with five different user-control strategies and the dependent variable was the cognitive load scores conducted following the completion of the learning process.

From the one-way ANOVA test, Levene's test for homogeneity of variances was not significant ($p > 0.05$), and therefore, the data do not violate the assumption of equality of error variances. A one-way ANOVA indicated there was significant difference on cognitive load across different control strategies, $F\ (4,260) = 10.64$, $p < 0.05$, *partial eta squared* $= 0.14$, with high effect size according to Cohen's 1998 guidelines (Pallant, 2007) as shown in Table 3.

The cognitive load mean scores indicated that students in the LUC mode obtained the lowest mean score ($M = 38.12$, $SD = 12.45$) followed by RUC mode ($M = 41.97$, $SD = 11.64$), FUC mode ($M = 45.72$, $SD = 11.22$), CUC mode ($M = 50.47$, $SD = 12.83$) and PC mode ($M = 51.18$, $SD = 13.19$), as shown in Table 4. To further analyze the results obtained, the Bonferroni test was carried out for between groups comparison study.

The results of the Bonferroni test indicated significant difference in terms of mean scores of cognitive load between LUC and FUC ($MD = 7.60$, $p < 0.05$) with LUC obtaining a lower mean score, a significant difference between LUC and CUC ($MD = 12.38$, $p < 0.05$) with LUC obtaining a lower mean score, a significant difference between LUC and PC ($MD = 13.06$, $p < 0.05$) with LUC obtaining a lower mean score, a significant difference between RUC and PC ($MD = 9.22$, $p < 0.05$) with RUC obtaining a lower mean score, and a significant difference between RUC and CUC ($MD = 8.51$, $p < 0.05$) with RUC obtaining a lower mean score. The results of the Bonferroni test also indicated no significant difference between LUC and RUC ($MD = 3.84$, $p > 0.05$), RUC and FUC ($MD = 3.75$, $p > 0.05$), PC and FUC ($MD = 5.47$, $p > 0.05$), and PC and CUC ($MD = 0.71$,

$p > 0.05$). This showed that LUC obtained the lowest mean score and it was significantly different with FUC, PC, and CUC, followed by RUC which was significantly different with PC and CUC. Meanwhile, there appears to be no significant difference in terms of mean scores between LUC and RUC.

4. Discussions

4.1. Research question 1

The overall findings indicated significant difference in post-test achievement between groups that undergo different user-control strategies of segmented animation. The pairwise comparison clearly shows that there is no significant difference between RUC, LUC and PC strategies. Hence, there appears to be significant difference between RUC and LUC strategies with FUC and CUC strategies. Moreover, students in RUC, LUC and PC strategies seem to have outperformed students in FUC and CUC strategies, in terms of adjusted mean scores of post-test. These findings signify that the segmented animation with user-control, either linear or random is more helpful than continuous animation strategy. The finding is consistent with Moreno (2007), which concluded that the segmented animation with linear user-control is more effective than continuous animation. The finding of Ahmad Zamzuri (2013) also shows that the segmented animation with linear user-control strategy is more superior to free user-control and continuous animation strategies. Further, the study by Mayer et al. (2003) also proved the effectiveness of the segmented animation with linear user-control strategy compared to the continuous animation display without control. Spanjers et al. (2010) also concluded statistically that the segmented animation with the linear user-control strategy is more effective in improving students' achievement compared to the continuous animation display without control. Although these studies are limited in comparing the segmented animation with continuous animation display, it is clear that segmented animation that is viewed in linear sequence with any user-control strategies is still more effective than the continuous animation display.

In detail, RUC is a user-control strategy that allows students to view the segmented animation linearly or any segment at random as needed by clicking the particular segment's button. While, in LUC strategy students can only display the segments in sequence by clicking the next and previous buttons. PC strategy is similar to LUC, the main differences are the pause duration between segments and the movements from one segment to another in linear were determined by the instructional developer. Similarly, of all these strategies involve linear movement from one segment to another. Linear movement seemed to encourage students to explore the entire animated content systematically without skipping the learning content in each segments. Even in RUC strategy, in addition to the linear movement, students were provided with freedom in choosing segments at random, but this freedom did not show any additional benefits to learning. Accordingly, the question arises whether students essentially take advantage of the additional feature available in this strategy throughout the learning process? The abstract contents might be the possible reason that discouraged students to use this feature effectively in the learning process. Future studies that explore the interaction of students with user-control features are necessary to answer this question, specifically in learning abstract topics.

Segmented animations that are displayed in linear form are seen to have an advantage in terms of providing students ample time to extract and interpret meaningful information systematically throughout the learning process, especially in learning abstract contents. This condition could overcome the attrition of information that would assist in the formation of meaningful mental models in the working memory for effective learning (Ahmad Zamzuri, 2013; Hasler et al., 2007). Segmented animation seems to also help in promoting active and passive processing in the students' memory structure. Whereby, active processing occurs during the animated display while passive processing occurs when the animation is in the paused condition between segments. The combination of dynamic and static displays potentially could overcome the over-confidence problem that might exist if the content is viewed just dynamically, as outlined by Awan and Stevens (2006).

In contrast to the continuous display characteristic as in the FUC and CUC strategies, the non-stop dynamic information entering the memory structure might cause the dropout of useful information due to the limited capacity and storage duration of the working memory. This shortcoming is not beneficial for the formation of meaningful mental model of

Table 1. Summary of tests between-subjects effect (RQ1).

Source	Df	Ms	*F*	Sig.	Partial eta Squared
Pre	1	2170.38	15.99	0.00	0.06
Group	4	813.19	5.99	0.00	0.09
Error	259	135.73			

Note. R Square = .11 (Adjustment R square = .10)

Table 2. Summary of descriptive statistic (RQ1).

		Unadjusted		Adjusted	
Student groups (control strategies)	*N*	*M*	SD	*M*	SE
RUC	57	67.88	1.82	67.75	1.56
LUC	49	67.28	1.68	67.33	1.67
PC	55	61.75	1.59	61.76	1.57
FUC	53	60.58	1.62	60.71	1.60
CUC	51	58.33	1.79	58.86	1.67

Note. Covariates appearing in the model are evaluated at the following value: Pre = 23.55

Table 3. Summary of tests between-subjects effect (RQ2).

Source	Df	Ms	*F*	Sig.	Partial eta Squared
Group	4	1604.16	10.64	0.00	0.14
Error	260	150.73			

Note. *R* Square = .14 (Adjustment *R* square = .13)

Table 4. Summary of descriptive statistic (RQ2).

		Cognitive load scores	
Student groups (control strategies)	*N*	*M*	SD
RUC	57	41.97	11.64
LUC	49	38.12	12.45
PC	55	51.18	13.19
FUC	53	45.72	11.22
CUC	51	50.47	12.83

the topic learned and likely contributes to the failure of effective learning, especially if plentiful learning time was not allocated. Even though students were provided freedom to stop and play the animation in FUC strategy, this feature does not help learning in comparison if the segments are determined by the instructional developer. This might be due to the inability of students in determining the meaningful stop points, since the topic learned is abstract. Further studies that explore students' interaction with the user-control features can answer this question. Finally, CUC strategy does not appear to be a good strategy for learning abstract contents in animation form. This strategy seems incapable of allocating appropriate time for students which would cause distractions and consequently, the role of animation as a teaching aid could fail (Mayer et al., 2003; Moreno, 2007; Spanjers et al., 2011).

4.2. Research question 2

The overall findings indicated significant difference in terms of cognitive load scores between groups that undergo different user-control strategy of segmented animation. Based on the findings, there are no significant differences in cognitive load scores between students in LUC and RUC, as well as RUC and FUC strategies. However, the cognitive load scores of students in LUC and RUC strategies are lower than FUC, PC and CUC strategies. Furthermore, it was found that there was no significant difference between FUC and PC strategies, as well as FUC and CUC strategies. This finding shows that the segmented animation in any user-control design, either linear, random, or free, have the potential in reducing the degree of cognitive load compared to the continuous animation display strategies.

Animation that changes over time possibly will increase the cognitive load during the viewing process (Ayres et al., 2009; Lowe, 2004). This is more apparent if the learning content is abstract and students face difficulty in depicting it visually. Limited working memory capacity, both in terms of capacity and duration, appeared to be major contributors to this situation (Sweller, 1994). This has potential in impeding students' ability to concentrate and extract adequate information from the animation presented for meaningful mental model formation in the working memory. Students' inability in processing new information and at the same time recalling and integrating earlier information simultaneously is a major factor contributing to cognitive over-load in the working memory (Ayres et al., 2009). Segmented animations with user-control features may potentially overcome this problem. Segmented animations with user-control features will allocate appropriate time for students to extract enough information to be processed in working memory effectively, specifically novices.

User-control strategies have been seen to have an effect in increasing cognitive load. It was found that the cognitive load scores for LUC strategy are lower than RUC strategy, followed by FUC strategy, although it was not significant. In LUC strategy, students only need to focus on the next and previous buttons. Likewise for RUC strategy, however, in RUC strategy students can also choose the segments randomly, as an additional user-control feature. Whereas in FUC strategy, students need to focus on three control buttons, which are the play, pause, and replay buttons. Split attention between the control buttons and animations' content might have an impact in increasing students' cognitive load in this study, although it was not significant. More control features could potentially divert students' attention to the operation of controls over the content of the lesson (Boucheix & Schneider, 2009). A further study that examines the impact of students' interaction with the control buttons on the cognitive load is necessary to reinforce this assumption.

Next, the cognitive load mean score of CUC strategy followed by PC was the highest compared to other strategies. Although PC strategy is in the form of segmented animation display, it is still controlled by program. An interactivity factor does not exist in this strategy. This may cause students to lose focus and fail to extract enough information from the relentless display of information to be processed in the working memory. With imperfect information, the load of working memory will certainly exist for the construction of a perfect mental model.

5. Conclusions

The findings showed that segmented animation courseware with linear or random user-control are more effective strategies in comparison with free user-control, program control and continuous user-control strategies. The findings also indicated that segmented animation with appropriate control strategies is capable in addressing the information dropout phenomenon in learning. Linear or random user-control strategies appeared to assist students in allocating appropriate time for the sensory memory to extract information from an animation display to be processed in the working memory before being registered in the long-term memory on a more-or-less permanent basis. Both of these strategies were also found helpful in improving students' achievement. In addition to achievement, both of these strategies also have a lower cognitive load mean score compared to other strategies. Cognitive load outcome also shows a notable relationship to students' achievement. To further affirm this, a regression analysis was conducted to identify the relation between these two variables. From the analysis it was found that cognitive load appeared to have significant relation with achievement ($p = 0.00$) with negative β value ($\beta = -0.37$). This finding indicates that increase in cognitive load will result decrease in achievement. Nevertheless, further study is needed to look on this issue in more detail. In summary, this study recommends LUC and RUC strategies as the superior user-control design in controlling segmented animation display in courseware, particularly for abstract topics.

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